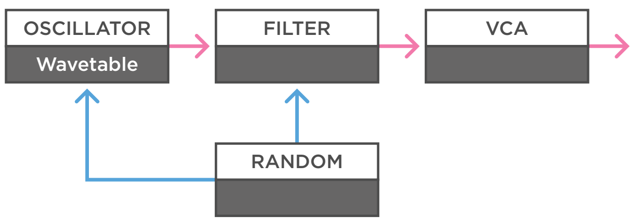
# Wavetable Oscillators

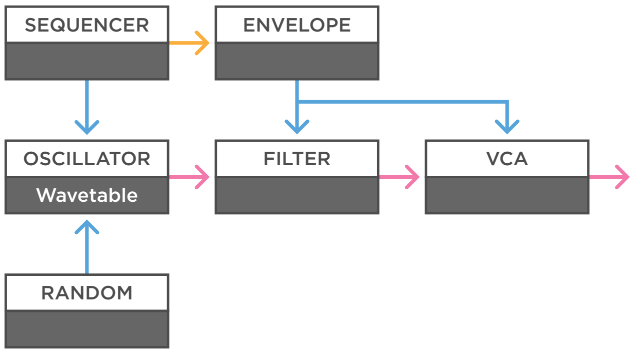
## Patch-tips



Audio
Control Voltage
Trigger / Gate




(a)




(b)

a - Let's start with a simple patch. When you use a wavetable oscillator in a very simple oscillator, filter & VCA setup, you can already achieve unique sounds. Because you can pick a complex waveform out of the many wavetables, it's easy to create sounds that are really hard to make with a normal oscillator. Then, if you use some modulation, like an LFO or smooth random voltage, you can use that to influence the filter or waveshape of the oscillator. This works great on low drone sounds.

b - The same concept also works on fast higher arpeggios, adding slow changes to a steady melody. If you add a simple envelope and sequencer to the patch you can create more controlled short melodic plucky sounds.

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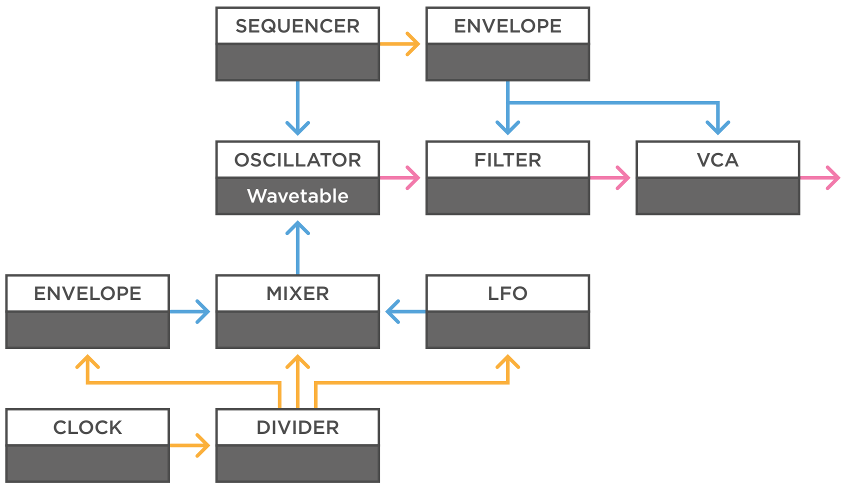
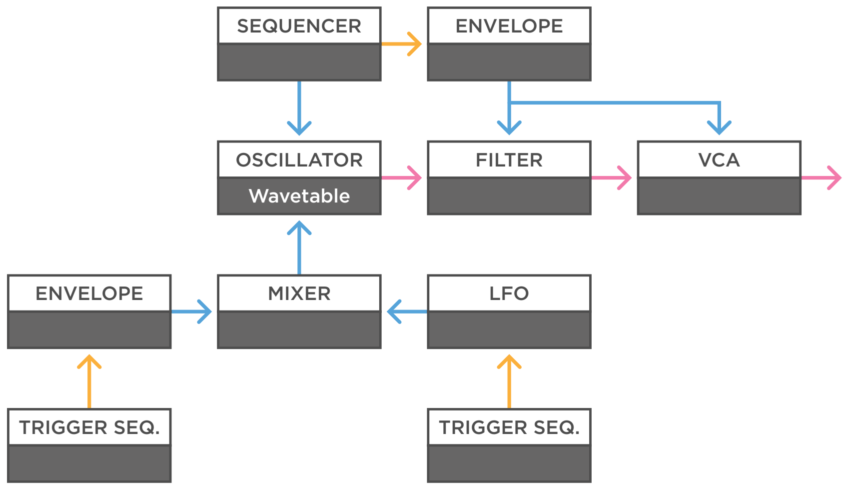



(a)




(b)

a - If you don't like the random aspect you can exchange the random voltage for something more controlled. An interesting trick is to send a clock source to a clock divider, and have different divisions trigger simple attack decay envelopes, or retrigger LFOs. If you send the envelopes and LFOs to a mixer and then to the waveshape input of the oscillator, you can create more complex, but tempo synced, rhythmic modulation in the waveshape.

When you're using a clock divider that outputs gates, you can use one of the slower divisions directly to the mixer to add a change that lasts a few bars.

b - If you want to have more control over the rhythm, you can replace the clock and divider with a couple of pattern sequences. The Beatstep Pro, for example, is a powerful tool for that. This way it's easy to program, change, mute and unmute the different patterns, which is great for performing.

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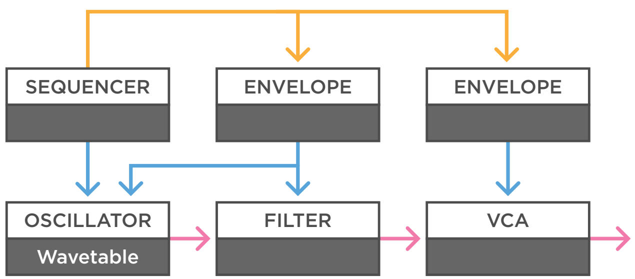
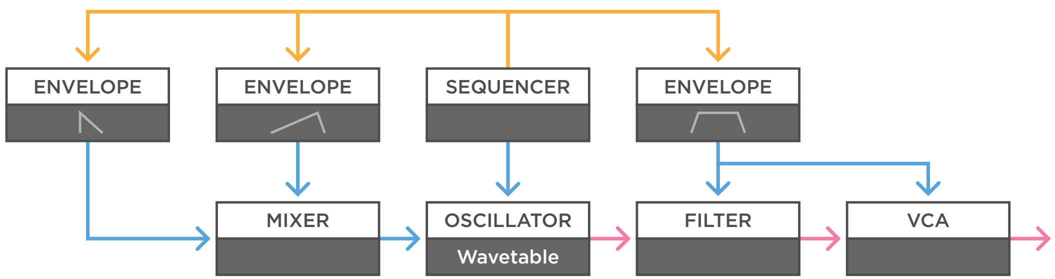



(a)




(b)

a - A common thing to do in a simple voice is to use two envelope generators, one to the filter, and one to the VCA. This way you have independent control over the filter movement, and the volume of the sound. You could gate the envelopes with a gate from a keyboard or sequencer, and to spice it up, multiply one of the envelopes to the waveshape.

b - You can also use a sequencer to gate an attack hold release envelope, which is sent to the filter and VCA, so the sound stays audible if you hold a note for a longer time. And at the same time use that same gate, on two other attack decay envelopes. One with a very fast attack and short decay, to create a little bit of movement in the transient of the sound. And one with a very slow attack, that changes the sound if you hold a note for a longer time. Then mix those envelopes together and send them to the waveshape input on the oscillator.

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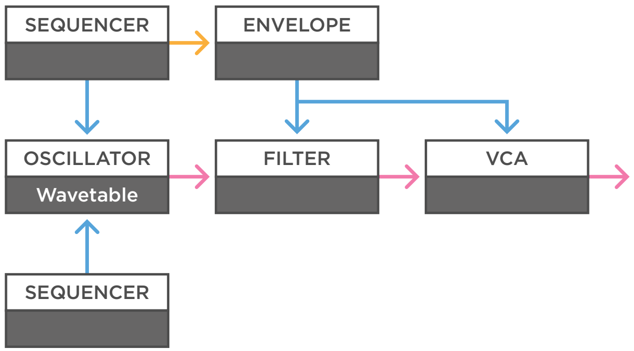
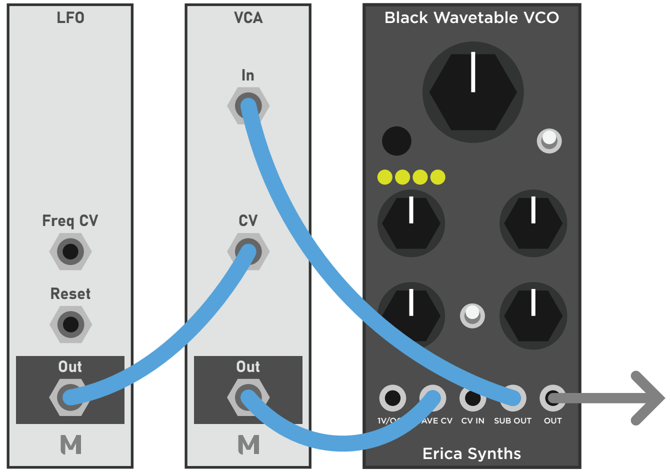



(a)




(b)

a - A simpler, but still very dynamic way to bring changes to your sound, is sending the output of a sequencer to the waveshape. This way you can dial in a different waveform for each step. This works great if you use different sequencers for the melody and waveshape, and set them to different lengths.

b - An easy trick with wavetable oscillators is to take one of its unused outputs, or a copy of the main output, and use it to modulate itself. For example, use it on the frequency modulation or waveshape input. This trick can lead to entirely new and interesting waveforms.

If you like to use this in a more dynamic way, you can place a VCA between the output and input of the wavetable oscillator. This way you can have something like a slow LFO control the amount of self-modulation.

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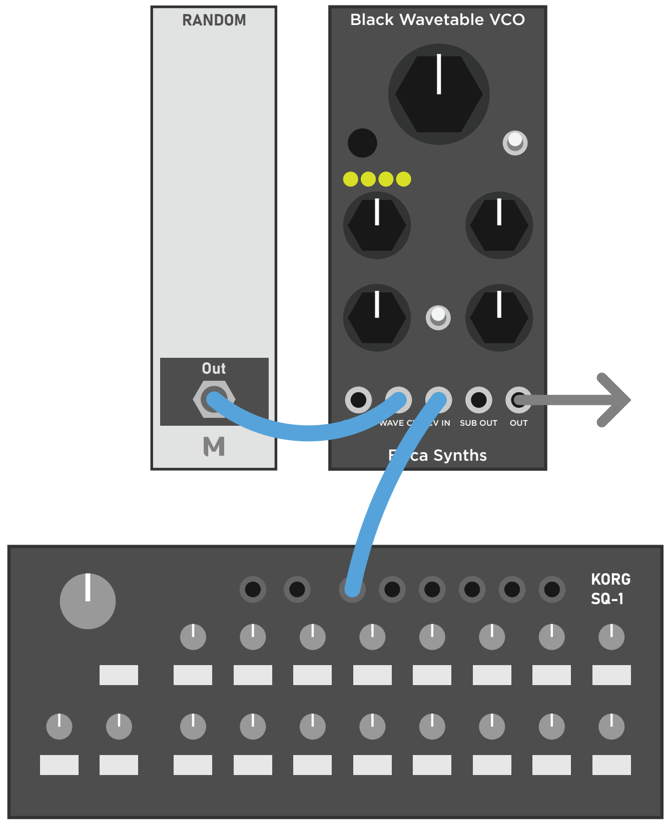


(b)

a - It's also worth it to experiment with waveshapes, by sending the output of a wavetable oscillator through a wavefolder. This works especially great if you mix different outputs of a wavetable oscillator together, before going to the folder.

Because the amount of the second output in the mix makes such a big difference in this patch, it's a great idea to add a VCA between the second output and the mixer, before the wavefolder. This way you can control the sound with a LFO or envelope.

b - The final tip is to experiment with the bank select input on a wavetable oscillator, if there is one. For example, use a slow smooth random voltage on the waveshape input, and at the same time switch between different banks with a CV sequencer.

Of course, you can also just send a smooth random voltage to the bank select input, to create absolutely unpredictable glitches and mayhem.